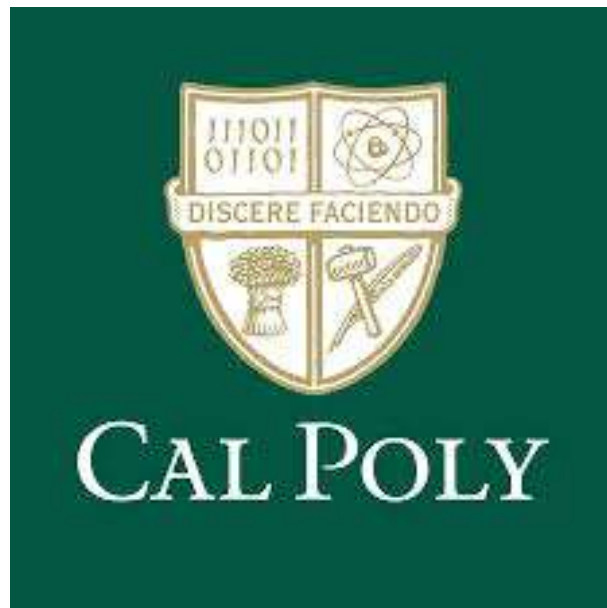




Access Control
&
Video Surveillance
Standard Guidelines

1 PURPOSE

1.1 Document Intent

Due to the complexity of access control systems and the variety of departments within CAL POLY affected, this document has been created. The Access Control Standard Guidelines provide guidelines and system intent that would not be covered in a project specification. These guidelines are meant to assist the project managers, consultants, contractors and CAL POLY parties with the installation, maintenance and management of access control systems.

This document is not intended to replace project drawings and project specifications. Project drawings and specifications will incorporate the information contained in this document as needed.

The Access Control & Video Surveillance Standard Guidelines is a living document that will be modified and updated as needed. A revision date is provided so that appropriate reference can be made.

2 GENERAL SYSTEM CRITERIA

2.1 General Overview

2.1.1 Terms Overview – Access Control vs. Security

The term “access control” and the term “security” are not interchangeable related to this document.

“Access Control” defines a system that restricts access to a facility based on a set of parameters. Access control systems include card reading devices of varying technologies and evidentiary cameras.

“Security” defines a system that includes active monitoring of a facility and includes active monitoring devices such as glass break devices on windows, horns on exit doors, and monitoring cameras.

This document covers the Access Control and Video Management standards at CAL POLY.

2.1.2 Types of Access Control Systems

2.1.2.1 Overview

Access control is divided into two categories: external perimeter building access control and interior access and security.

Each building achieves perimeter access control through a Honeywell ProWatch system including card readers. External perimeter access control is maintained via **building time schedules**.

Interior access control and security is determined by the needs of the individual schools, departments, and staff on a building by building basis.

2.1.2.2 Internal Access Control & Video Monitoring

Internal access control and security can be achieved in several ways and is driven by the needs of the occupants. Security measures range from monitoring doors being held open to adding cameras to monitor personnel entering and exiting the space. Security measures will add additional cost to the project and will be designed on a case by case basis.

Depending on the size and complexity of the project and the security needs, a separate Security Consultant may be engaged aside from the Security Contractor to provide for an engineered solution.

2.1.2.3 External Perimeter Access Control & Video Monitoring

The philosophy of the external access control system is to provide for the open campus feel during the day and to provide for a secure campus during nights and weekends.

External perimeter access control is achieved through a Honeywell system with both campus wide components and local building components. All new access control systems for both existing buildings as well as new buildings will meet the standards outlined in these guidelines.

Equipment and installation as outlined in these guidelines will make up the access control and video monitoring system.

A select number of card reader devices and cameras are available with the Honeywell system. This system is installed per university standards for Honeywell systems by an approved university vendor that is certified by Honeywell to install and program this equipment and software, as outlined in these guidelines.

Programming of the system is performed by a university approved vendor. This system is installed per university standards for Honeywell systems by an approved university vendor that is certified by Honeywell to install and program this equipment and software, as outlined in these guidelines.

Naming convention for locations (buildings, rooms, etc.) Shall follow a predefined University Standard.

Badging and the assignment of access levels for card holders is performed by XXXXXXXXXXXXXXXXXXXX.

Maintenance of the system is the responsibility of XXXXXXXXXXXXXXXXXXXX. A university approved vendor is necessary to work on the system. Campus Lock Shop can assist with system maintenance and warranty issues as needed.

Depending on the installation and scope of the project, design and commissioning may be created by a Security Consultant separate from the Security Contractor. The complexity of the system and the amount of devices will dictate if a Security Consultant should be part of the project team.

2.1.2.2.1 Honeywell ProWatch & MaxPro Systems

Additional equipment can be purchased to extend the Honeywell ProWatch & MaxPro system for the External Perimeter or Internal Access Control of each building and to extend the video monitoring system.

A select list of card readers and cameras are available with the Honeywell system for extension of the systems. This system is installed per university standards for Honeywell systems by an approved university vendor as outlined in these guidelines.

Programming of the system is performed by a university approved vendor.

Credentialing of the system and the assignment of access levels for card holders is performed by XXXXXXXXXXXXXXXX.

Maintenance of the system is the responsibility of the department/school. A university approved vendor is necessary to work on the system. XXXXXXXXXXXX can assist with system maintenance and warranty issues as needed.

A list of university approved Honeywell security contractors can be found in Appendix B.

2.1.2.4 Legacy Systems

Those buildings on campus, running on other systems shall be phased out and replaced with Honeywell ProWatch, MaxPro Video Management and EquiP cameras on a project by project basis.

2.1.2.5 Intrusion Systems

Intrusion alarms may also be used for security purposes. These systems, when installed with the appropriate equipment, can be connected to the ProWatch System. The Control Panels and Arming Stations shall be Honeywell VISTA-128FBP, Vista 128BPE, Vista 128BPT, VISTA-250FBP, Vista 250BPE, or Vista 250BPT Controllers.

1. Features of the ProWatch Interface:
2. Disarm and unlock a door on card swipe.
3. Arm and lock a door on card swipe.
4. Common area arm/disarm.
5. Access denied if intrusion system is in alarm or armed.
6. Monitor and log intrusion system events and alarms in ProWatch
7. Associate intrusion system events & alarms to video surveillance integrations.

2.1.3 Device Locations

Generally, access control and video monitoring devices, such as card readers and cameras, are located on the exterior of buildings or spaces. Card readers are also used to limit access into interior space. By Fire Code, card reader placement cannot be located in areas that “trap” people and limit egress of a

building or space.

Panels and recording devices are located inside the building in a appropriately secure space. ACS Panels and NVR recording devices must be provided in spaces not accessible by the public and in some cases, require card reader secured locations.

Network recording devices are located either within a building in a secure, climate controlled space, or can be located remotely in the main data center.

2.1.3.1 Responsibilities

All devices shall be located by design based on the needs of the building occupants, site visits and drawing reviews. Review and confirmation of the locations is provided by XXXXXXXXXXXXXXXXXXXX through review of the access control drawings.

Locations of IP cameras shall be coordinated through XXXXXXXXXXXXXXXXXXXX as part of the Security scope of the project.

Locations of E-phones shall be coordinated through XXXXXXXXXXXXXXXXXXXX.

2.2 Project by Scope

The scope of work of each project and the involvement of trades will depend on the size of the project related to access control.

For projects with larger scopes, including new buildings, major renovations that include the external doors of an existing building, and in some cases change of use of a building, a Security Consultant is to be part of the consulting team.

For projects with smaller scopes, including minor door changes to existing buildings, an assessment is provided by UPD and FMS and/or Consultants and contractors as engaged by the University.

2.2.1 Large Scope Projects

For projects with large scopes of work, as defined above, the drawings for a project follow those of the project consultants. Architectural backgrounds are provided to the Security Consultant and a full scope of drawings and specifications for the project are provided. The Security Consultant will follow the phases of the lead consultant and will have drawings included in the standard schematic design, design documents, and construction documents phases. The Security Consultant will review submittals and perform construction administration duties as well, as per contract.

Once brought on board, the Security Consultant will meet with the Users and the PM to assess the needs of the facility. The initial drawings will be reviewed by the University Access Control team, which includes XXXXXXXXXXXXXXXXXXXX, for compliance. The full drawings set will then proceed along with the lead consultant's and general contractor's schedules.

The Security Consultant will develop a full permit set of drawings and specifications for bid to the Security Contractor.

2.2.2 Small Scope Projects

For projects with a smaller scope of work, as defined above, FMS and UPD act as the primary initial point of contact (POC). FMS and UPD work with the User and the PM to assess the needs and determine a scope of work.

In cases where the scope of work dictates a full set of drawings and specifications, FMS and UPD will alert the PM as to the need to hire a Security Consultant.

In all other cases, FMS and UPD will turn the assessment into a bid package and will take the work to bid with the Security Contractor.

2.2.2.1 Commissioning

All access control and security system installations must be commissioned. Commissioning includes the testing of a complete system from the card reader device, for example, to the database. The commissioning process is included in the scope of work by both the Security Consultant and the Security Contractor. A system is accepted as complete only after the commissioning process is complete.

2.2.3 Security Contractor

The Security Contractor is hired to install the designed access control and security system. The Security Contractor is responsible for acquiring the necessary hardware and equipment, pulling all wires, and making all final system connections. The Security Contractor is also responsible for coordination with XXXXXXXXXXXXXXXXXXXX to program the devices of the system and ensure a complete and working system is delivered. All services to be provided by the Security Contractor are part of the security specifications.

Additionally, the Security Contractor is responsible for functional testing of the system once installed and for all adjustments that need to be made after the system has been commissioned.

2.2.3.2.1 Requirements

Security Contractors must be approved by the university. [Insert Approval Procedure here](#)

At minimum, the Security Contractor shall hold the legally required California State Contractor's licenses necessary to accomplish the installation and the Security Contractor shall also hold current company certifications to work on the ProWatch and MaxPro System by Honeywell.

2.2.3.2.2 Approved List

[See Appendix B for approved vendor list.](#)

2.3 System Overview

2.3.1 System Philosophy

The intent of the access control system is to provide for a controllable perimeter of a building that can fit the needs of the building occupants. All exterior doors shall be electronically controlled. This allows for additional flexibility especially during emergency situations.

Some doors are equipped with the ability to enter and exit all the time using card readers. Some doors are equipped with the ability to enter and exit only during certain times. Some doors are for emergency exit only and are equipped with monitoring devices. For devices required per door type, see Section 4.

2.3.2 Electronic Access Control System (EACS)

In general, the EACS comprises of a wide-area data network, enterprise server, enterprise administrative and monitoring clients, communications client, remote system controllers, and card readers compliant with the campus credentialing system.

The existing EACS system is a Honeywell ProWatch Corporate Edition.

2.3.3 Badging System

The credentialing system is designed to provide credentials for faculty, staff, students, support and administrative personnel. The system is managed by [XXXXXXXXXXXXXXXXXX](#).

2.3.4 Video Surveillance System (VSS)

The digital surveillance system, MaxPro VMS and NVR is manufactured by Honeywell. The system is LAN network-based and uses digital video servers,

network recording media, and video management software.

The VSS shall be integrated with the access control system, Honeywell ProWatch Access Control System.

2.4 Document Requirements

2.4.1 Drawings and Specifications

Regardless of the size and scope of the project, drawings and specifications shall be included and followed. For smaller projects, hand sketches and sheet specs are acceptable. For larger projects, full CAD drawings and book specs are required.

2.4.2 Submittals

As the specifications indicate, all projects regardless of the size and scope shall include submittals of all devices and wiring included on the project. Submittals shall be provided to the PM and/or general contractor for tracking and approval. No wiring or devices shall be installed prior to written approval of a submittal.

2.4.3 As-built Record Drawings

As the specifications indicate, all projects regardless of the size and scope shall include requirements for as-built record drawings at the end of each project. For smaller projects, hand drawn red-lines may be accepted. For large projects, only CAD as-builts will be accepted.

Close out documents should include all equipment manuals and information provided by the manufacturer. Documentation shall also include electronic versions of the panel schedules and wire diagrams

As part of the close out document package, the security contractor shall include as-built version of the consultant documents. These updated documents will include, but are not limited to:

- Door Schedules
- Floor Plans
- Door Details
- Camera Details
- Equipment Cabinet Details
- Door Wiring Schematics
- Single Line Diagrams
- Cabinet Configuration
- Functional Diagram
- Panel Wiring Diagram

Documentation shall include any changes made in the field to the system as shown in the construction drawings and shall include conduit and wire runs for future reference.

Documents shall be provided in both PDF and CAD formats. CAD files shall not be PDF files referenced into a .dwg file. CAD files shall be properly referenced and shall include the color plot table.

3 COORDINATION

3.1 Coordination with CAL POLY

3.1.1 FMS

Facilities Management Services (FMS) is the university department responsible for the operations and maintenance of all university buildings.

Related to access control, FMS is responsible for the maintenance of the hardware of the system. FMS determines the standards for installation, including writing of the doors and door frames, types of wires used, and mounting of equipment.

FMS reviews all phases of the access control documentation for compliance with installation standards, doors and hardware coordination, and equipment locations as related to the existing building.

3.1.2 UPD

The University Police Department (UPD) is a group with the primary goal to provide a safe and secure environment in which the social and academic endeavors of the university community may be fully realized.

Related to access control, UPD is responsible for assessing and approving the locations of all access control devices. UPD determines the feasibility of the location of equipment and advises on the needs related to equipment. UPD is responsible for monitoring and responding to all university access control and security issues.

UPD review all phases of the access control documentation for compliance with device locations and standard equipment.

3.1.3 ITS

Information Technology Services (ITS) is the university department responsible for the installation and maintenance of all networks.

Related to access control, ITS is responsible for providing network connectivity to

the building and access control devices, as well as the maintenance of this connectivity. ITS determines the standards for all networks and network devices.

ITS reviews all phases of the access control documentation for compliance with network and network device standards. ITS makes recommendations if a building network needs to be upgraded as well.

3.1.4 Card Services

Card Services are managed by the ~~XXXXXXXXXX~~ department whose responsibility is to facilitate identity for fast, reliable, secure, convenient access to university provided products and services.

Related to access control, CAL POLY Card is responsible for the creation of the credentialing database including the issuing and assigning of all user rights and privileges, as directed by the customer, as well as all access cards.

4 PRODUCT STANDARDS AND REQUIREMENTS

4.1 Standard Card Reader Devices for New Buildings & Existing Buildings

For a new access control system in any new building on campus, the standard card reader will be a multi-technology, contactless reader. The following models are the only acceptable devices.

For mullion mounting applications: Honeywell OM17BHONSP

For typical wall mounting applications: Honeywell OM42BHONSP

4.3 Standard Request to Exit (REX) Devices

4.3.1 For New Buildings

For a new building, request to exit devices shall be included in the exit device hardware.

See Section 5 – Installation Standards by Opening Type for more detail.

4.3.2 For Existing Buildings

For existing buildings, request to exit devices should be included in the exit device hardware. Additional types of REX devices can be discussed on a project by project basis.

See Section 5 – Installation Standards by Opening Type for more detail.

4.4 Standard Electrified Hardware

4.4.1 For New Buildings

In general, mortise locks and exit devices are acceptable for new buildings. Electrified hinge, transfer pivot, or a continuous hinge with EPT is specified depending on the type of door and door devices.

See Section 5 – Installation Standards by Opening Type for more detail.

4.4.2 For Existing Buildings

For an existing building, devices shall follow standards as much as possible. Openings and requirements can be reviewed on a project by project basis.

See Section 5 – Installation Standards by Opening Type for more detail.

4.5 Standard Honeywell Equipment – Field Hardware

The security management system shall be equipped with access control field hardware required to receive alarms and administer all access granted/denied decisions. All field hardware shall meet UL requirements.

PW6K1IC	PW-6000 Intelligent Controller 32 I/O and/or reader boards combined. Includes on-board Ethernet device.
PW6K1ICE	PW-6101 Single door PoE Intelligent Controller Supports one door using a single reader or two readers with an in/out reader configuration. Provides 2 inputs/2 outputs and up to 16 PW6K1R1E reader modules.
PW6K1R1E	PW-6101 Single door PoE Reader Module to be used with the PW6K1ICE or PW6K1IC Supports one door using a single reader or two readers with an in/out reader configuration. Provides 4 inputs/2 outputs.
PW6K1IN	PW-Series 16 Input Module - 16 general purpose input relays
PW6K1OUT	PW-Series 16 Output Module - 16 general purpose output relays
PW6K1R2	PW-Series Two Reader Module
PW5K1MX8	Series 8-Port Multiplexer
PWEP1502	Mercury Controller with 2 reader ports

4.5.1 Security Terminal Cabinets

STCs can be comprised of either a Hoffman enclosure or a Honeywell PW-Series

Controller Enclosures. STCs must be lockable. The lock/key type shall follow CAL POLY standards for these types of enclosures.

STCs shall only be located in electrical rooms, telecom rooms or IDF rooms. Locations shall be coordinated ahead of time. Rooms such as janitor's closets and hidden or concealed rooms are not acceptable locations.

4.6 IP Cameras

New EQUIP Micro Dome Camera	
H2W2GR1	Network TDN Low-Light, 120db Wide Dynamic, IR Micro Dome Camera, 1/2.8" Imager, 2MP, Video Analytics, 2.8 mm Fixed Lens, IR LEDs, PoE, H.265+/H.265/H.264, ONVIF S/G/Q, UL-CAP
New EQUIP Indoor Dome Camera	
H3W2GR1V	Network TDN Low-Light, 120db Wide Dynamic, IR Indoor Dome Camera, 1/2.8" Imager, 2MP, Video Analytics, 2.7-13.5 mm MFZ Lens, 3 IR LEDs, PoE, H.265+/H.265/H.264
H3W2GR2	Network TDN Low-Light, 120db Wide Dynamic, IR Indoor Dome Camera, 1/2.7" Imager, 2MP, 7-22 mm MFZ Lens, 3 IR LEDs, PoE, H.264
New EQUIP Outdoor/Rugged Dome Camera	
H4L2GR1V	Network TDN Ultra Low-Light, 140db Wide Dynamic, IR Rugged Dome Camera, 1/2.8" Imager, 2MP, Video Analytics, Open VA framework, 2.7-12 mm MFZ Lens, 3 IR LEDs, PoE, H.265+/H.265/H.264
H4W2GR1V	Network TDN Low-Light, 120db Wide Dynamic, IR Rugged Dome Camera, 1/2.8" Imager, 2MP, Video Analytics, 2.7-13.5 mm MFZ Lens, 3 IR LEDs, PoE, H.265+/H.265/H.264
H4W4GR1V	Network TDN Low-Light, 120db Wide Dynamic, IR Rugged Dome Camera, 1/3" Imager, 4MP, Video Analytics, 2.7-13.5 mm MFZ Lens, 3 IR LEDs, PoE, H.265+/H.265/H.264
H4D8GR1	Network TDN Low-Light, Digital Wide Dynamic, IR Rugged Dome Camera, 1/1.7" Imager, 12MP, 4.1-12.8 mm MFZ Lens, 3 IR LEDs, PoE, H.265/H.264
New EQUIP IR Bullet Cameras	
HBL2GR1V	Network TDN Ultra Low-Light, 140db Wide Dynamic, IR Bullet Camera, 1/2.8" Imager, 2MP, Video Analytics, Open VA framework, 2.7-12 mm MFZ Lens, 4 IR LEDs, PoE, H.265+/H.265/H.264
HBW2GR1V	Network TDN Low-Light, 120db Wide Dynamic, IR Bullet Camera, 1/2.7" Imager, 2MP, Video Analytics 2.7-13.5 mm MFZ Lens, 4 IR LEDs, PoE, H.265+/H.265/H.264
HBW4GR1V *	Network TDN Low-Light, 120db Wide Dynamic, IR Bullet Camera, 1/3" Imager, 4MP, Video Analytics, 2.7-13.5 mm MFZ Lens, 4 IR LEDs, PoE, H.265+/H.265/H.264

HBW2GR3V	Network TDN Low-Light, 120db Wide Dynamic, IR Bullet Camera, 1/2.7" Imager, 2MP, Video Analytics, 5-60 mm MFZ Lens, 4 IR LEDs, PoE, H.265+/H.265/H.264
HBL6GR2	Network TDN Low-Light, Digital Wide Dynamic, IR Rugged Dome Camera, 1/1.8" Imager 6MP, Video Analytics, 4.1-16.4mm MFZ Lens, 4 IR LEDs, PoE, H.265/H.264
HBD8GR1	Network TDN Low-Light, Digital Wide Dynamic, IR Bullet Camera, 1/7" Imager, 12MP, 5.1-12.8 mm MFZ Lens, 4 IR LEDs, PoE, H.265/H.264
New EQUIP IR 360° Fisheye lens Cameras	
HFD6GR1	Network TDN 6MP IR Fisheye, 1/1.8" Imager, 1.7mm fixed lens, 3 IR LEDs, PoE, H.265/H.264
HFD8GR1	Network TDN 12MP IR Fisheye, 1/1.7" Imager, 1.98mm fixed lens, 3 IR LEDs, H.265/H.264
New EQUIP Multi-Sensor Cameras	
HM4L8GR1	Network Low Light, Digital Wide Dynamic, 1/1.9 " 2MP CMOS (4X), IR Rugged Dome Camera, 4* 5mm/M12 fixed lens, PoE+, H.265/H.264
HMBL8GR1	Network Low Light, Digital Wide Dynamic, 1/2.8 " 2MP CMOS (4X), IR Rugged Bullet Camera, 4*3mm/M12 fixed lens, PoE+, H.265/H.264
New EQUIP Series Camera Mounts and accessories	
HBC5WT	Wall/Ceiling Camera Mount, 5", White
HD3-MK2	Indoor Mini-Dome Wall Mount Bracket
HD4CHIP-WK2	Rugged Dome Wall Mount Bracket
HD4CHIP-PK2	Rugged Dome Pendant Mount Bracket
HB4G-PM	Pole Mount Adapter
HB34G-CM	Corner Mount Adapter
HBG-BB	Bullet Camera Back Box
HFG-WK	Fisheye Wall Mount Bracket
HFG-PK	Fisheye Pendant Mount Bracket

4.6.2 NVR/DVR

Record and monitor up to 64 IP channels at 1920 fps @ 4CIF/VGA or 1920 fps @ 720p HD or 1280 fps @ 1080p (4 Mbps bitrate) HD per MAXPRO NVR SE. Network bandwidth/throughput supported per NVR with Incoming: 160 Mbps, Outgoing: 350 Mbps providing a total: 510 Mbps. Archival support of 16 channels @ 4 Mbps bitrate each (64 Mbps total archival throughput capacity per MAXPRO NVR SE), with outgoing archival storage throughput of 200 Mbps. Multi-stream support with maximum 128 streams per MAXPRO NVR SE. Support for configuring one (1) preferred stream for continuous recording and one (1) preferred stream for live video/motion based recording per camera. Support for One-Way Audio (for specific

IP cameras) with live, playback and clip export on NVR desktop client for up to 64 IP channels.

Storage capacity shall allow for at minimum 30 days based upon the following recording standards: 11FPS recording on 50% motion, using the H.265 or H.264 compression codec.

4.6.3 ProWatch Client Workstation

Honeywell ProWatch Video Client software allows for monitoring, surveillance, and review capabilities and functionality at the client workstations. At minimum, ProWatch Client software shall be provided on all projects. Additionally, some projects may include the physical workstation in the specifications, if an existing workstation is not available.

PRO-WATCH CLIENT WORKSTATION¹ REQUIREMENTS

FEATURE	REQUIREMENT
Recommended Processor	Intel® Core™ i3, i5, or i7 Processor 2.8 GHz or higher.
Recommended Operating System	Windows® 8/8.1 (32 or 64 bit) Windows 10 Windows 2012/2012/2014 Server Enterprise or Standard edition (32 or 64 bit)
Recommended PC Type	Workstation or Server hardware
Recommended System Memory (RAM) ¹	4 GB for Windows 7/8/8.1 (32 and 64 bit) 4 GB for Windows 10
DVD Drive	Yes
Hard Drives	Capacity, Speed, Interface: 80 GB or larger 7,200 RPM SATA or SAS
USB connection	2.0
Network connection	100Mbps/sec or greater, GB recommended
Video resolution	1024x768 pixels; 24 bit color or higher
Multimedia	Standard VGA Graphics Adapter (Display adapter)

¹ Includes RAM required by the client machine to run the Windows operating system.

4.6.4 MaxPro Client Viewer Workstation

MaxPro Client is capable of live viewing of up to 64 IP cameras on a single remote workstation with up to two (2) monitors set up at CIF resolution. For 4CIF and HD resolution, the number of live streams needs to be benchmarked based on client hardware configuration deployed. Cost-effective enhanced HD video rendering on remote desktop clients with support for monitoring of up to 23 1080p HD cameras in real time (30 fps)/690 fps 1080p HD with no-time lapse using the GPU capabilities of in-built processor graphics with 6th generation Intel® Core™ Processors for client systems. Up to 4 1080p HD @ 20 fps/80 fps on local client.

Enhanced Monitoring with Desktop Clients

Multi-zoom views on HD video and support for Profile cameras to create virtual cameras by digitally zooming into the field of view. Example: Zoom in on a cash register in one view of the HD camera while at the same time monitoring the cash operator in the zoom out view of the HD camera.

Ability to investigate events and alarms by simultaneously viewing alarm videos at various stages. For every alarm, users can view the video captured during pre-alarm, on-alarm, and post-alarm, and also view live video from the camera which triggered the alarm.

Enriched video viewing experience through the intuitive video rendering engine that optimizes CPU utilization by altering the video frame rate.

Cost-effective enhanced H.264 and H.265 HD video rendering on remote desktop clients with support for monitoring of up to 23 1080p HD cameras in real time (30 fps)/690 fps 1080p HD with no-time lapse using the GPU capabilities of in-built processor graphics with 6th generation Intel® Core™ Processors for client systems. Up to 4 1080p HD @ 30 fps/120 fps on local client.

Support for auto switching of streams utilized by the client with multi-stream cameras, saving workstation hardware cost by displaying more cameras simultaneously with lower resolution streams instead of the primary HD streams

Cameras will be set up to utilize:

Surrounding Cameras: Provide option to grant a user the ability to view a single camera surrounded by the cameras programmed as the “Surrounding Cameras”.

3D Positioning: The user shall be able to view a specific object in the live video in a 3-dimensional view. This feature is available in the Context menu options and is only supported with New equip PTZs HDZ302DE, HDZ302D, and HDZ302DIN. 3D positioning options include: Click-based camera positioning, Rectangle selection 3D positioning, and Restore to last PTZ position.

Profile Cameras: Multi-zoom views on HD video and support to create virtual cameras by digitally zooming into the field of view. For example: zooming in on a cash register in one view of the HD camera while at the same time monitoring the cashier in the zoomed out view of that HD camera.

4.7 Standard Wire Requirements

4.7.1 Cable Acceptable for Reader, Door contacts, REX and Power

This cable includes:

- (1) 18 AWG, 4 conductors PVC/PVC unshielded - Grey jacket
- (1) 22 AWG, 4 conductors PVC/PVC unshielded - Blue jacket
- (1) 22 AWG, 4 conductors PVC/PVC unshielded - White jacket
- (1) 22 AWG, 6 conductors PVC/PVC overall shielded - Orange jacket

4.7.2 Cable Acceptable for Cameras

The preferred cable to be used for cameras include

- (1) 24 AWG, 4p, CAT6 plenum rated

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4.8.1 Device Network Connections

One Ethernet connection is required at the system controller location.
Each NVR requires an IP connection.
Each Ethernet location requires an individual IP address.

4.9 Power Requirements

4.9.1 Emergency Power

Where available, access control devices shall be provided optional standby power from a generator. If generator power is not available, battery back-up shall be included.

4.9.2 Power for Electric Unlocking

Doors with electric unlocking or latch pull back require one dedicated circuit.

4.9.3 Power for Mortise Locks

Mortise locks should be provided with 12 volts. For locations of Mortise locks, see Section 5 – Installation Standards by Opening Type.

4.10 Specialty Areas

4.10.1 Elevators

4.10.1.1 Card Access to Enter Elevator with Open Floor Access

4.10.1.2 Card Access to Enter Elevator with Controlled Floor Access

4.11.2 Roll Up Doors

4.11.3 Parking Lots

5 INSTALLATION STANDARDS BY OPENING TYPE

5.1 Entrances with Card Readers

5.1.1 General Location

Entrances with Card Readers are designated as the main location for access to a building during off hours and on weekends. Typically these doors are specified

near a parking area or walkway for ease of accessibility. These doors may also be specified close to elevators and ADA accessible entrances.

At least one entrance location is required to have a card reader on the exterior of a building. For the interior, this location would be the main entrance location to a secure space.

5.1.2 Equipment

An Entrance with Card Reader is equipped, at minimum, with a card reader, a request to exit (REX) device, electrified hardware, and a door contact.

5.1.3 Doors and Hardware

5.1.3.1 Exterior - Single Door - With Mortise Lock

For a single exterior door with a card reader and a Mortise lock, specifications shall include the request to exit (REX) device in the Mortise lock and shall fail secure. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Cylindrical locks, electric strikes and door loops are not acceptable.

5.1.3.2 Exterior - Single Door - With Exit Device

For a single exterior door with a card reader and an exit device, specifications shall include the request to exit (REX) device in the exit device and shall include electric latch retraction. The power transfer hinge shall be an electrified continuous hinge with EPT or a transfer pivot.

Electric strikes and door loops are not acceptable.

5.1.3.3 Exterior – Pair of Doors - With Mortise Lock

For a pair of exterior doors with a card reader and a Mortise lock, specifications shall include the request to exit (REX) device in the Mortise lock and shall fail secure. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Cylindrical locks, electric strikes and door loops are not acceptable.

5.1.3.4 Exterior – Pair of Doors -With Exit Device – Rim Devices with Mullion

For a pair of exterior doors with a card reader and rim devices with mullion, specifications shall include the request to exit (REX) device in the exit device. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Electric strikes and door loops are not acceptable.

5.1.3.5 Exterior – Pair of Doors -With Exit Device – Concealed Vertical Rod

For a pair of exterior doors with a card reader and concealed vertical rods, specifications shall include top and bottom latching and the request to exit (REX) device shall be in the exit device. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Less bottom rod, electric strikes, surface vertical rods and door loops are not acceptable.

5.1.3.6 Interior – Single Door - With Mortise Lock

For a single interior door with a card reader and a Mortise lock, specifications shall include the request to exit (REX) device in the Mortise lock. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Electric strikes and door loops are not acceptable.

5.1.3.7 Interior – Single Door - With Exit Device

For a single interior door with a card reader and an exit device, specifications shall include the request to exit (REX) device in the exit device and shall include electric latch retraction or electrified trim. The power transfer hinge shall be an electrified continuous hinge with EPT or a transfer pivot.

Electric strikes and door loops are not acceptable.

5.1.3.8 Interior – Pair of Doors- With Mortise Lock

For a pair of interior doors with a card reader and a Mortise lock, specifications shall include the request to exit (REX) device in the Mortise lock. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Electric strikes and door loops are not acceptable.

5.1.3.9 Interior – Pair of Doors- With Cylindrical Lock

For a pair of interior doors with a card reader and a cylindrical lock, specifications shall include the request to exit (REX) device in the cylindrical lock. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Electric strikes are not acceptable.

5.1.3.10 Interior – Pair of Doors - With Exit Device – Concealed Vertical Rod

For a pair of interior doors with a card reader and concealed vertical rods, specifications shall include the request to exit (REX) device shall be in the exit device. Less bottom rods are acceptable. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Electric strikes and surface vertical rods are not acceptable.

5.1.4 Wire Requirements

5.1.4.1 Type of Wire

Entrances with a Card Reader must be provided with the approved composite cable as per Section 3.2.

5.1.4.2 Conduit

Conduit is not required for inside the door frame as the wire is plenum rated. A junction box shall be provided in the ceiling above the door for wire access. Conduit may be necessary from the junction box back to the location of the Honeywell panel based on the type of ceiling in the building.

5.1.4.3 Color Scheme

The following color schedule shall be used:

White – Door Contacts

Blue – Request to Exit (REX)

Orange – Card Readers

Grey – Locks

5.2 Normal Operations Entrances

5.2.1 General Location

Normal Operations Entrances are designated as any door that can be accessed during normal business hours, but is not useable for entrance during off-hours. Normal Operations Entrances are electronically locked during off hours and on weekends for entrance but can always be used as an exit.

Normal Operations Entrances would consist of the group of doors without card readers and those not designated as Exit Only Entrances.

5.2.2 Equipment

A Normal Operations Entrance is equipped, at minimum, with a request to exit (REX) device, a door contact, and electrified hardware.

5.2.3 Doors and Hardware

5.2.3.1 Exterior - Single Door - With Mortise Lock

For a single exterior door with a Mortise lock and without a card reader, specifications shall include the request to exit (REX) device in the Mortise lock and shall fail secure. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Cylindrical locks, electric strikes and door loops are not acceptable.

5.2.3.2 Exterior - Single Door - With Exit Device

For a single exterior door with an exit device and without a card reader, specifications shall include the request to exit (REX) device in the exit device and shall include electric latch retraction. The power transfer hinge shall be an electrified continuous hinge with EPT or a transfer pivot.

Electric strikes and door loops are not acceptable.

5.2.3.3 Exterior – Pair of Doors - With Mortise Lock

For a pair of exterior doors with a Mortise lock and without a card reader, specifications shall include the request to exit (REX) device in the Mortise lock and shall fail secure. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Cylindrical locks, electric strikes and door loops are not acceptable.

5.2.3.4 Exterior – Pair of Doors - With Exit Device – Rim Devices with Mullion

For a pair of exterior doors including rim devices with mullion and without a card reader, specifications shall include the request to exit (REX) device in the exit device. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Electric strikes and door loops are not acceptable.

5.2.3.5 Exterior – Pair of Doors - With Exit Device – Concealed Vertical Rod

For a pair of exterior doors with concealed vertical rods and without a card reader, specifications shall include top and bottom latching and the request to exit (REX) device shall be in the exit device. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Less bottom rod, electric strikes, surface vertical rods and door loops are not acceptable.

5.2.3.6 Interior – Single Door - With Mortise Lock

For a single interior door with a Mortise lock and without a card reader, specifications shall include the request to exit (REX) device in the Mortise lock. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Electric strikes and door loops are not acceptable.

5.2.3.7 Interior – Single Door - With Exit Device

For a single interior door with an exit device and without a card reader, specifications shall include the request to exit (REX) device in the exit device and shall include electric latch retraction or electrified trim. The power transfer hinge shall be an electrified continuous hinge with EPT or a transfer pivot.

Electric strikes and door loops are not acceptable.

5.2.3.8 Interior – Pair of Doors - With Mortise Lock

For a pair of interior doors with a Mortise lock and without a card reader, specifications shall include the request to exit (REX) device in the Mortise lock. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Electric strikes and door loops are not acceptable.

5.2.3.9 Interior – Pair of Doors - With Cylindrical Lock

For a pair of interior doors with a cylindrical lock and without a card reader, specifications shall include the request to exit (REX) device in the cylindrical lock. The power transfer hinge shall be an electrified hinge or transfer pivot.

Electric strikes are not acceptable.

5.2.3.10 Interior – Pair of Doors - With Exit Device – Concealed Vertical Rod

For a pair of interior doors with concealed vertical rods and without a card reader, specifications shall include the request to exit (REX) device in the exit device. Less bottom rods are acceptable. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Electric strikes and surface vertical rods are not acceptable.

5.2.4 Wire Requirements

5.2.4.1 Type of Wire

Normal Operations Entrances should be provided with the approved composite cable as per Section 3.2.

5.2.4.2 Conduit

Conduit is not required for inside the door frame as the wire is plenum rated. A junction box shall be provided in the ceiling above the door for wire access. Conduit may be necessary from the junction box back to the location of the Honeywell panel based on the type of ceiling in the building.

5.2.4.3 Color Scheme

The following color schedule shall be used:

White – Door Contacts

Blue – Request to Exit (REX)

Orange – Card Readers

Grey – Locks

5.3 Exit Only Entrances

5.3.1 General Location

Exit Only Entrances are designated as any door that can be used only as a means of egress regardless of the time of day. Exit Only Entrances do not have door hardware or key ways to be able to be entered from the outside.

Exit Only Entrances are typically designated only on the outside of a building. Examples of an Exit Only Entrance include exits from stairwells and large auditoriums.

5.3.2 Equipment

A door indicated as an Exit Only Entrance is equipped, at minimum, with door contacts and electrified hardware.

5.3.3 Doors and Hardware

5.3.3.1 Exterior - Single Door - With Mortise Lock

For a single exterior exit only door with a Mortise lock, specifications shall include the request to exit (REX) device in the Mortise lock. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Cylindrical locks and door loops are not acceptable.

5.3.3.2 Exterior - Single Door - With Exit Device

For a single exterior exit only door with an exit device, specifications shall include the request to exit (REX) device in the exit device. The power transfer hinge shall be an electrified continuous hinge with EPT or a transfer pivot.

Door loops are not acceptable.

5.3.3.3 Exterior – Pair of Doors - With Mortise Lock

For a pair of exterior exit only doors with a Mortise lock, specifications shall include the request to exit (REX) device in the Mortise lock. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Cylindrical locks and door loops are not acceptable.

5.3.3.4 Exterior – Pair of Doors - With Exit Device – Rim Devices with Mullion

For a pair of exterior exit only doors including rim devices with mullion, specifications shall include the request to exit (REX) device in the exit device. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Electric strikes and door loops are not acceptable.

5.3.3.5 Exterior – Pair of Doors - With Exit Device – Concealed Vertical Rod

For a pair of exterior exit only doors with concealed vertical rods, specifications shall include top and bottom latching and the request to exit (REX) device shall be in the exit device. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Less bottom rods, surface vertical rods and door loops are not acceptable.

5.3.3.6 Interior – Single Door - With Mortise Lock

For a single interior exit only door with a Mortise lock, specifications shall include the request to exit (REX) device in the Mortise lock. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Door loops are not acceptable.

5.3.3.7 Interior – Single Door - With Exit Device

For a single interior exit only door with an exit device, specifications shall include the request to exit (REX) device in the exit device. The power transfer hinge shall be an electrified continuous hinge with EPT or a transfer pivot.

Door loops are not acceptable.

5.3.3.8 Interior – Pair of Doors - With Mortise Lock

For a pair of interior exit only doors with a Mortise lock, specifications shall include the request to exit (REX) device in the Mortise lock. The power transfer hinge shall be an electrified hinge or a transfer pivot.

Door loops are not acceptable.

5.3.3.9 Interior – Pair of Doors - With Cylindrical Lock

For a pair of interior exit only doors with a cylindrical lock, specifications shall include the request to exit (REX) device in the cylindrical lock. The power transfer hinge shall be an electrified hinge or transfer pivot.

5.3.3.10 Interior – Pair of Doors - With Exit Device – Concealed Vertical Rod

For a pair of interior exit only doors with concealed vertical rods, specifications shall include the request to exit (REX) device in the exit device. Less bottom rods are acceptable. The power transfer hinge shall be an electrified continuous hinge with EPT or transfer pivot.

Surface vertical rods are not acceptable.

5.3.4 Wire Requirements

5.3.4.1 Type of Wire

Exit Only Entrances should be provided with the approved composite cable as per Section 3.2.

Optionally, individual wires for the devices required at the door can be specified but the color scheme must be adhered to.

5.3.4.2 Conduit

Conduit is not required for inside the door frame as the wire is plenum rated. A junction box shall be provided in the ceiling above the door for wire access. Conduit may be necessary from the junction box back to the location of the Honeywell panel based on the type of ceiling in the building.

5.3.4.3 Color Scheme

The following color schedule shall be used:

White – Door Contacts

Blue – Request to Exit (REX)

Orange – Card Readers

Grey – Locks

6 ITS REQUIREMENTS AND STANDARDS

6.1 Network Device Registration

All network devices must be registered on the CAL POLY network. All devices with MAC addresses, including but not limited to, the NVR, cameras, video servers, and Honeywell Intelligent Controller boards, need to be registered with ITS to receive an IP address.

6.1.1 Registration Procedure

The following registration procedures shall be used for all access control devices requiring IP addresses.

- Contractor shall provide a list of devices and MAC address to the Project Manager.
- Project Manager shall send an email to XXXXXXXXXXXXXXXXXXXX indicating each device and MAC address.
Included in the email, shall be the contact information designated as:
- Project Manager shall ensure that a list of devices, MAC addresses, and IP addresses is included in the project close out documents.

7 PROGRAMMING REQUIREMENTS AND STANDARDS

7.1 Programming

The security contractor is responsible for programming to ensure a complete working system and integration between the existing Honeywell system and the new components. Programming rights are provided by XXXXXXXXXXXX on a project by project basis.

Requests for programming rights should be made in writing to the Project Manager at least **2 days** prior to the need. The Project Manager shall make arrangements with XXXXXXXXXXXX to have access available.

7.2 Device Naming and Coordination

Each device and door shall have a name consistent with CAL POLY naming standards. Once the design is complete, and all equipment has been coordinated, XXXXXXXXXXXX will use the construction documents to create names.

7.2.1 CAL POLY Responsibility

XXXXXXXXXXXX will provide the Access Control Equipment Template in an Excel electronic version. Information that will be completed by XXXXXXXXXXXX includes:

7.3 Time Schedule Set Up

The security contractor will set up a default time schedule as part of the initial system programming. The general schedule will have the doors open from 7am to 7pm. Alternate schedules shall be coordinated between UPD, the security contractor, the building occupants and the Project Manager.

8 Card Access

CAL POLY Card Services manages and maintains the credentialing system and operates independent of the access control system.

The Project Manager is responsible for communicating with CAL POLY Card Services to coordinate user access and door schedules prior to the system go-live.

8.1 Time Schedules

Maintenance of the time schedule and changes??

8.2 User Access

All user access requirements, including re-carding if necessary, may be completed by CAL POLY Card Services. However, it is important to note the access program does not dictate carding requirements nor re-carding. The Project Manager shall coordinate conversations between the building occupants, UPD and CAL POLY Card Services.

9 Maintenance and Building Operations

9.1 System Monitoring

9.1.1 Doors

Active monitoring of all electrified doors is provided through the Honeywell system. During an alarm condition, such as a door being held open, an alert will be sent through the Honeywell system to UPD and any Client Workstations.

9.1.2 Cameras

The primary objective of cameras is to provide evidentiary footage if necessary. Some locations have active monitoring via Client Workstations determined by the needs of the building occupants.

10.2 System Problems

10.2.1 Card Readers and Door Issues

When issues occur with card readers and doors related to entry to a building, Lock Shop Should be called at **XXXXXXXXX**.

Plaques are posted at card reader doors providing the phone number for UPD as well as the device name for reference.

10.2.2 Other issues

APPENDIX A - FORMS

APPENDIX B - APPROVED VENDORS

APPENDIX C -

APPENDIX D -

APPENDIX E - EXAMPLE REQUIRED DIAGRAMS

APPENDIX F - DOOR HARDWARE INSTALLATION PHOTOS

APPENDIX G - EMERGENCY PHONES

APPENDIX X - CUTSHEETS

